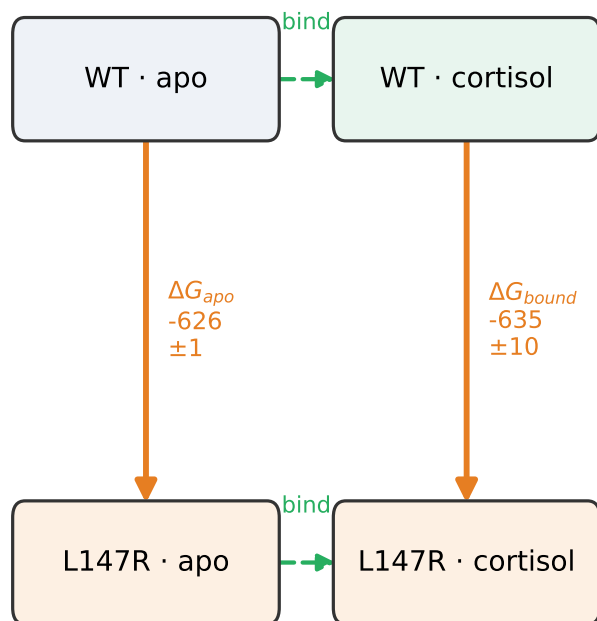


Tier-2 relative binding FEP (pmx + GROMACS non-equilibrium TI): L147R × cortisol — demonstration

A Alchemical thermodynamic cycle

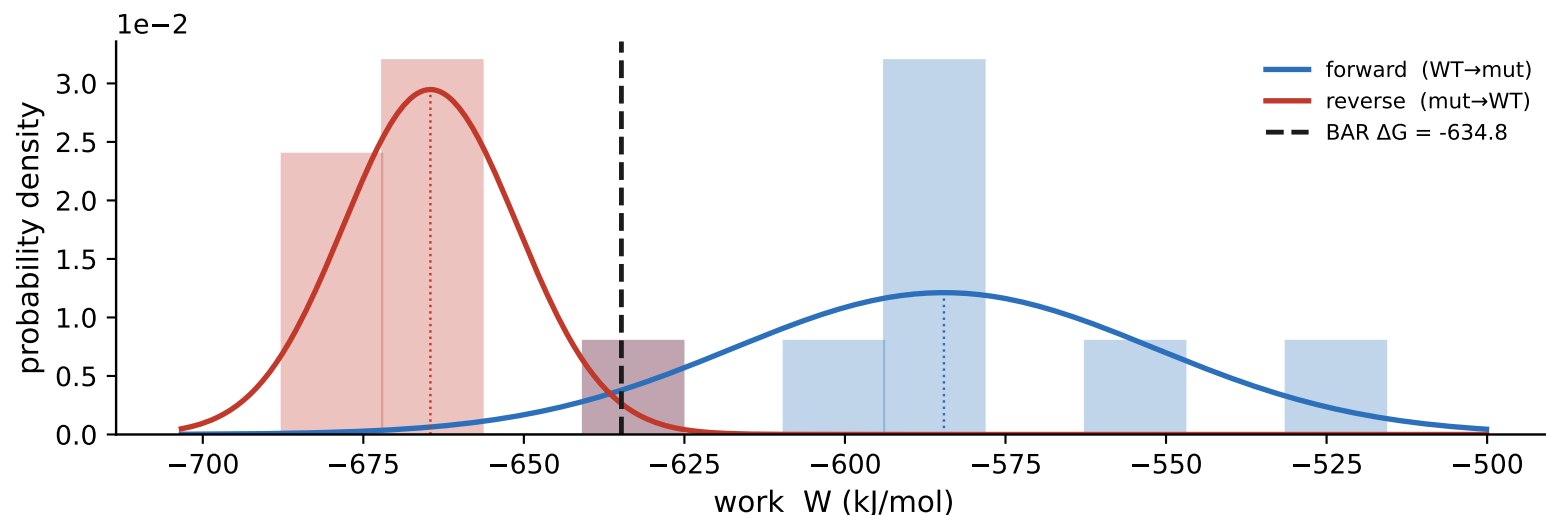


$$\Delta\Delta G_{bind} = \Delta G_{bound} - \Delta G_{apo}$$

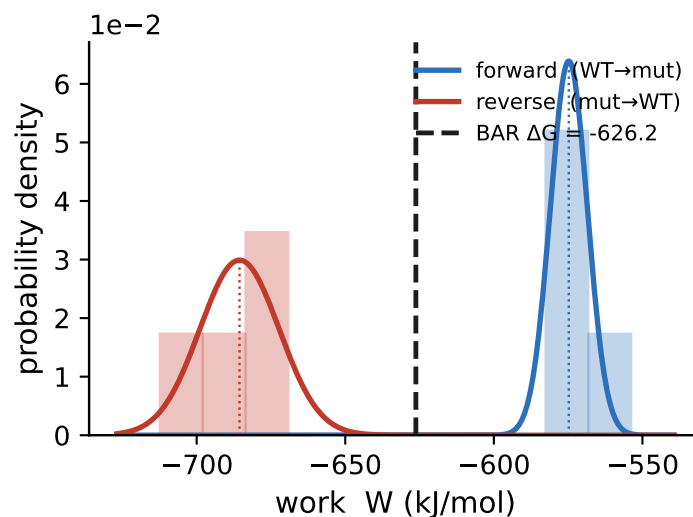
$$= -634.8 - (-626.2) = -8.6 \text{ kJ/mol}$$

WT → L147R morph on both legs; the common ~ -630 kJ/mol charge-insertion cost cancels in the difference

B Bound leg — non-equilibrium work (8+8 transitions)



C Apo leg



D L147R favors cortisol

